EMISSION AND PRICE OPTIMIZATION IN ENERGY PRODUCTION

M. Burak KORKMAZ
150220326

Selman Toker
150220330

Abdullah AYDOGAN
150230731

CONTENT

- Project Aim
- Datasets
- Objective Function
- Constraints
- Solution Space
- Algorithms
- Algorithm Comparison
- Conclusion

Project Aim

Our goal is to optimize the following objectives in France and Germany:

- **Minimizing Greenhouse Gas Emissions**
- **Minimize Electricity Prices**
- **Maintaining Total Energy Production**

DATASETS

eurostat

We utilized 3 datasets from EUROSTAT

- Total Energy Production by Energy Source and Year in European Countries
- Electricity Prices by Year in European Countries
- Greenhouse Gas Emissions by Year in European Countries

Objective Function

- x_i: Annual average electricity prices for household consumers (kw/h) in country i.
- y_i: Share of renewable energy in total energy production in country i.
- z_i: Greenhouse gas emissions from electricity generation for country i.

$$\min_{x,y,z} f(x, y, z) = \alpha x + \beta z - \gamma y$$

Constraints

Cost Constraint

Electricity price should not increase by more than 10% per year

$$x \leq 1.10 \cdot x_0$$

where x is the current electricity price and x_0 is the electricity price in the previous years.

Constraints

Emission Constraint

Greenhouse gas emissions should be reduced by at least 5 percent per year

$$z \leq 0.95 \cdot z_0$$

where z is the current greenhouse gas emissions and z_0 is the greenhouse gas emission in the previous years.

Constraints

Energy Constraint

Countries' energy production should remain stable

$$|T_{\text{year}+1} - T_{\text{year}}| \leq \delta$$

where T_{year} is the total energy production in the given year, and δ is the acceptable deviation.

Solution Space

- x_0: electricity price in the previous years
- z_0: greenhouse gas emission in the previous years
- T_0: total energy production in the previous year
- δ: acceptable deviation in total energy (tolerance)
- k: conversion factor from emissions to energy units

$$S = \left\{ (x, y, z) \in \mathbb{R}^3 \,\middle|\, \begin{array}{l} 0 \leq x \leq 1.10 \cdot x_0 \\ 0 \leq z \leq 0.95 \cdot z_0 \\ |y + k \cdot z - T_0| \leq \delta \end{array} \right\}$$

x: electricity price

y: renewable energy production

z: greenhouse gas emission

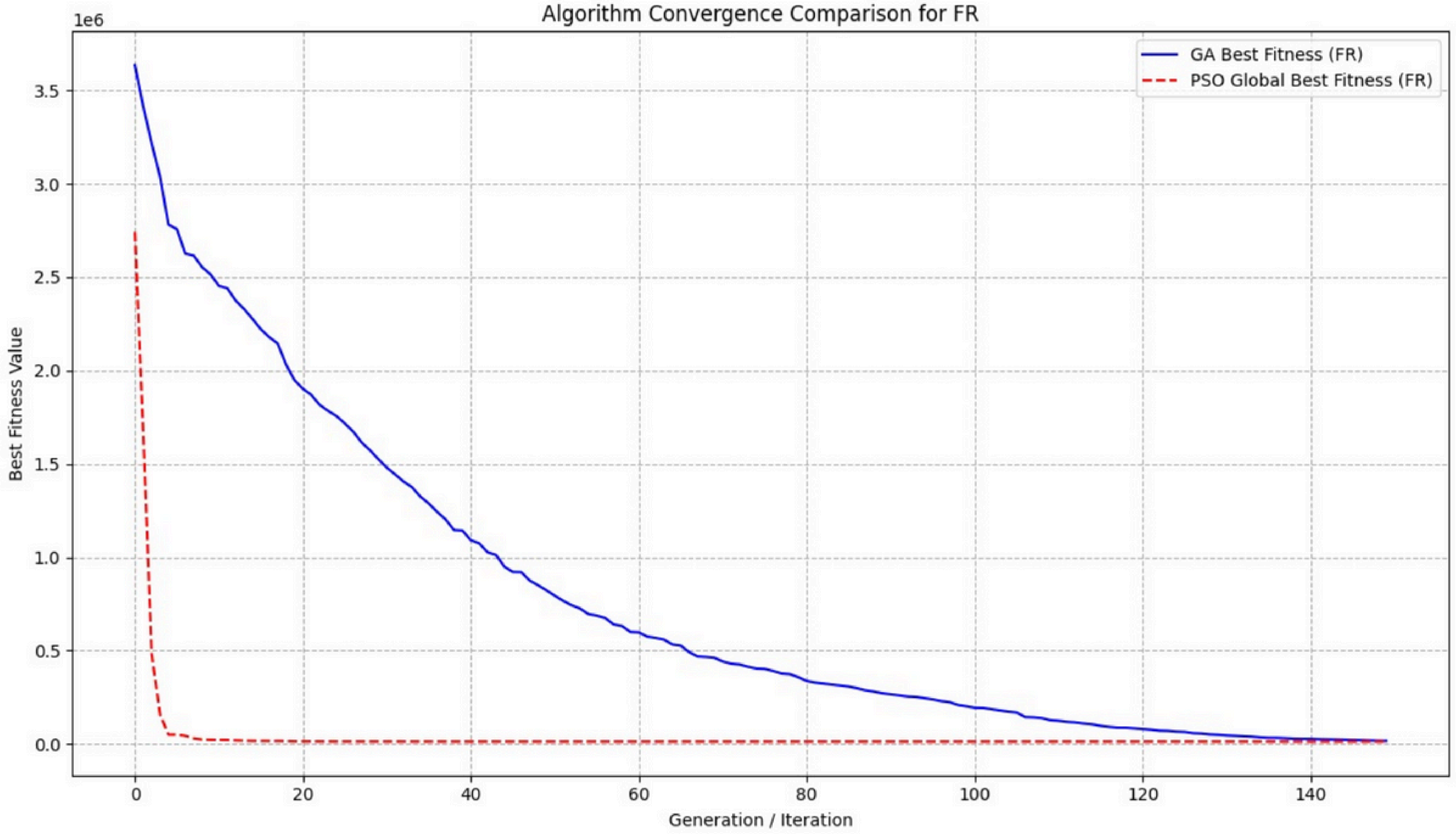
Algorithms

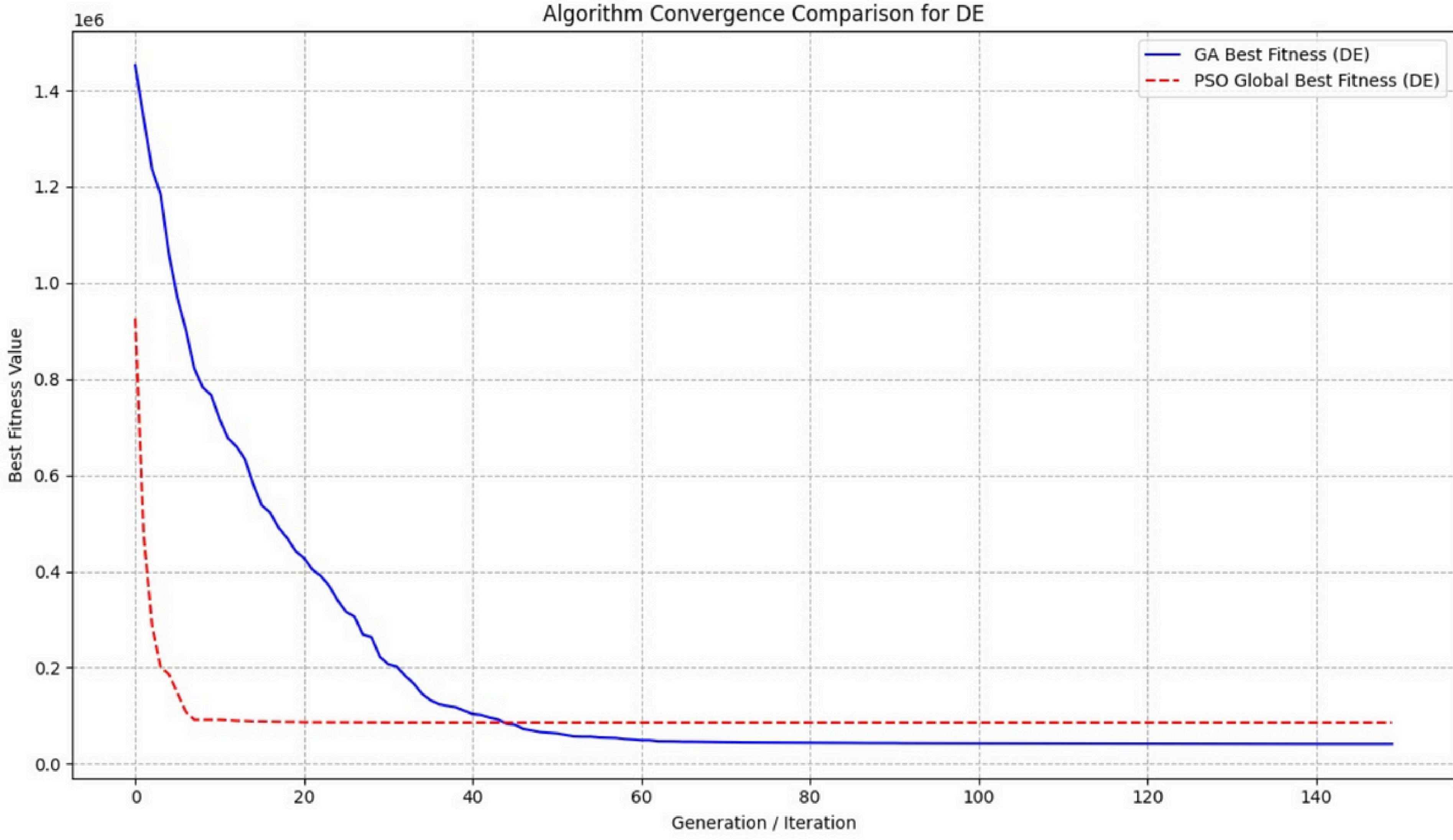
Genetic Algorithm

- POPULATION_SIZE_GA = 200
- MAX_GENERATIONS_GA = 300
- MUTATION_RATE_GA = 0.08
- CROSSOVER_RATE_GA = 0.85

PSO Algorithm

- NUM_PARTICLES_PSO = 100
- MAX_ITER_PSO = 300
- W_PSO = 0.5
- C1_PSO = 1.5
- C2_PSO = 1.5





Algorithm Comparison

Results of **Genetic** Algorithm for **Germany**

	Initial	**Final**
Price (€/kWh)	0.3318	0.3390
Renewable Share	38.32%	57.12%
CO2 Emission (million tone)	216.55	205.71

Results of **PSO** Algorithm for **Germany**

	Initial	**Final**
Price (€/kWh)	0.3318	0.3671
Renewable Share	38.32%	56.96%
CO2 Emission (million tone)	216.55	205.72

Results of **Genetic** Algorithm for **France**

	Initial	**Final**
Price (€/kWh)	0.2148	0.2230
Renewable Share	35.27%	39.39%
CO2 Emission (million tone)	25.53	24.26

Results of **PSO** Algorithm for **France**

	Initial	**Final**
Price (€/kWh)	0.2148	0.2406
Renewable Share	35.27%	45.45%
CO2 Emission (million tone)	25.53	23.47

Algorithm Comparison

Energy Shares Percentage in Best Found Solution in **France**

Sources	Initial	Final	Change
Nuclear_heat	43.82	47.11	3.29
Natural_gas	6.80	6.94	0.14
Hydro	6.77	10.07	3.30
Wind	5.68	8.95	3.27
Solar_photovoltaic	2.92	5.55	2.63
Bioenergy	1.55	3.83	2.28
Oil_and_petroleum_products_(excluding_biofuel_portion)	0.94	0.80	-0.14
Pumped_hydro_power	0.82	2.88	2.05
Primary_solid_biofuels	0.68	2.61	1.94
Other_bituminous_coal	0.62	0.63	0.02
Solid_fossil_fuels	0.62	0.57	-0.05
Fuel_oil	0.56	0.69	0.13
Biogases	0.45	1.65	1.21
Non_renewable_waste	0.33	0.00	-0.33
Non_renewable_municipal_waste	0.31	0.19	-0.12
Renewable_municipal_waste	0.31	1.68	1.36
Gas_oil_and_diesel_oil_(excluding_biofuel_portion)	0.31	0.27	-0.04
Blast_furnace_gas	0.25	0.00	-0.25
Manufactured_gases	0.25	0.26	0.00
Other_fuels_nec	0.08	0.00	-0.08
Tide,_wave,_ocean	0.07	1.69	1.62
Refinery_gas	0.06	0.00	-0.06
Industrial_waste_(non_renewable)	0.02	0.00	-0.02
Geothermal	0.02	1.69	1.67
Other_oil_products_nec	0.02	0.00	-0.02
Other_liquid_biofuels	0.00	1.66	1.66
Other_recovered_gases	0.00	0.28	0.28

Energy Shares Percentage in Best Found Solution in **Germany**

Sources	Initial	Final	Change
Solid_fossil_fuels	13.18	11.79	-1.39
Wind	9.14	14.27	5.13
Lignite	8.45	8.39	-0.06
Natural_gas	6.12	6.77	0.65
Solar_photovoltaic	4.47	9.49	5.02
Other_bituminous_coal	4.46	4.66	0.19
Bioenergy	3.63	8.56	4.93
Nuclear_heat	2.54	7.11	4.56
Biogases	2.36	6.75	4.40
Hydro	1.29	5.50	4.21
Primary_solid_biofuels	0.78	4.88	4.10
Manufactured_gases	0.70	0.89	0.19
Non_renewable_waste	0.47	0.00	-0.47
Blast_furnace_gas	0.46	0.61	0.15
Pumped_hydro_power	0.44	1.90	1.46
Non_renewable_municipal_waste	0.41	0.35	-0.07
Renewable_municipal_waste	0.41	1.89	1.48
Oil_and_petroleum_products_(excluding_biofuel_portion)	0.38	0.00	-0.38
Coking_coal	0.16	0.00	-0.16
Coke_oven_gas	0.16	0.00	-0.16
Other_oil_products_nec	0.15	0.00	-0.15
Gas_oil_and_diesel_oil_(excluding_biofuel_portion)	0.09	0.08	-0.01
Other_fuels_nec	0.09	0.00	-0.09
Fuel_oil	0.08	0.14	0.05
Other_recovered_gases	0.08	0.00	-0.08
Brown_coal_briquettes	0.07	0.00	-0.07
Industrial_waste_(non_renewable)	0.06	0.16	0.10
Refinery_gas	0.05	0.00	-0.05
Anthracite	0.04	0.06	0.02
Geothermal	0.02	1.90	1.88
Other_liquid_biofuels	0.01	1.95	1.94
Tide,_wave,_ocean	0.00	1.93	1.93

Conclusion

- We have managed to reduce CO_2 emissions to less than 5% without a major change in electricity prices.

- We have achieved more sustainable energy production by increasing the use of renewable energy sources.

THANKS FOR LISTENING